

SANJAY SAKTHIRAJAN

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EDUCATION

University of Minnesota - Twin Cities

Dec 2026

M.S., Data Science for Operations Research

• GPA: 3.87/4.00

PSG College of Technology

Jul 2025

B.E., Mechanical Engineering

• GPA: 3.17/4.00

CERTIFICATIONS

• Databricks Certified Data Engineer Associate

Sep 2026

• AWS Certified Cloud Practitioner (CLF-C02)

Jul 2026

TECHNICAL SKILLS

Data Engineering: Databricks, Apache Spark, PySpark, Spark SQL, Delta Lake, Delta Sharing, ETL/ELT, Medallion Architecture, Databricks Jobs

Cloud & AWS: AWS, Amazon S3, Lambda, Glue, EMR, DynamoDB, SNS, SQS, EventBridge

Governance & Quality: Unity Catalog, Data Quality, Data Reconciliation, Pipeline Observability, Schema Management

Programming & Databases: Python, SQL, MySQL, pandas, NumPy

DevOps & Testing: Git, GitHub, GitHub Actions, pytest, CI/CD Fundamentals

Analytics & Tools: Power BI, Excel, scikit-learn, XGBoost, CatBoost, PyCharm, Jupyter Notebook, Streamlit

PROJECTS

Healthcare Claims Lakehouse & Data Integration Platform

2026

- AWS S3 | Databricks | PySpark | Spark SQL | Delta Lake | Unity Catalog | Python | SQL | pytest | GitHub Actions
- Built a governed AWS S3 + Databricks lakehouse using Bronze/Silver/Gold Delta architecture for public/synthetic CMS healthcare data, with PySpark/Spark SQL transformations, DQ validation, reconciliation, and operational audit tables.
- Engineered explicit-schema Carrier pipelines processing 4.74M claims and normalized 13 repeating service-line groups into 8.85M service records while validating zero duplicate service-line keys.
- Productionized reusable Carrier and PDE pipelines, processing 5.55M prescription events; orchestrated Carrier -> PDE with Databricks Jobs and separated reusable Python transformations from execution notebooks.
- Implemented reusable DQ/audit utilities, 20 passing pytest tests, GitHub Actions CI, and Unity Catalog-governed S3 access through a storage credential and external location.

Invoice Data Extractor

2026

- Python | Pydantic | MySQL | Claude API | Streamlit
- Built an end-to-end pipeline converting unstructured PDF invoices into validated structured records using Python, schema-based Claude tool use, Pydantic models, and MySQL.
- Integrated invoice upload, extraction, relational persistence, and results visualization in a Streamlit application.

NBA Championship Predictor

2026

- Python | SQL | MySQL | NBA API | pandas | scikit-learn | Streamlit
- Built an end-to-end pipeline to collect and store 25 seasons of NBA team statistics from an API in MySQL, then transformed historical data into model-ready features for predictive analysis.
- Engineered performance metrics and trained classification models to estimate championship probabilities, integrating data ingestion, relational storage, feature engineering, evaluation, and Streamlit output.

Shot-Make Probability Technical Project - Basketball Operations Application

2026

- Python | pandas | scikit-learn | CatBoost | XGBoost | Git/GitHub
- Built and compared probability models on 425,719 training observations and generated validated predictions for 213,977 test observations using CatBoost, XGBoost, feature engineering, and log-loss evaluation.
- Validated probability bounds, row counts, shot-ID alignment, and submission integrity while maintaining a reproducible Git/GitHub workflow.

EXPERIENCE

Ganga Associates

Jun 2022 - Sep 2022

Operations Intern

- Analyzed production and workflow data using time studies, Excel, and Python, contributing to process improvements that increased line output from 20 to 22 units per shift.
- Built Power BI reports and Python-generated summaries to identify bottlenecks and communicate operational findings to production and engineering teams.

PSG Industries

Aug 2020 - Jun 2024

Industrial Trainee

- - Analyzed manufacturing performance data using Excel, Python, and pandas, and built XGBoost models in Databricks to predict bottlenecks, helping the team target process improvements and reduce cycle time.
- Developed process maps and operational documentation to improve visibility into manufacturing workflows.